



ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your *Hormones*, Ultimate Guide

Your Hormones Are Talking. Here Is How to Hear What They Are Actually Saying, and What to Do About the Message.

A guide to the root causes behind PCOS, low testosterone, and HPA-driven hormonal symptoms, for women and men who have been told their labs are "normal" and still do not feel like themselves.

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BEFORE YOU START

A note before you start. This guide is educational information about how sex hormones, stress hormones, and metabolic function interact. It is not individualized medical advice, it does not diagnose any condition, and it does not replace a relationship with your physician or a clinical evaluation with my team.

A few things worth stating plainly before you read further:

This guide does not treat hormonal birth control, testosterone replacement therapy (TRT), or metformin as the enemy, now or ever. If you take hormonal birth control, understand that a regular-seeming cycle on the pill is a withdrawal bleed created by the medication itself, not necessarily proof that the underlying pattern this guide discusses has resolved. That is simply how the medication works. It is not a reason to stop or change your birth control. If you are already on prescribed TRT, or your physician is discussing it with you, nothing here should be read as a reason to start, stop, or adjust that therapy on your own. If you have been prescribed metformin for insulin resistance or PCOS, keep taking it exactly as prescribed. The nutrient and lifestyle strategies in this guide are additive to these treatments, not a substitute for any of them, and any change to a prescription is a conversation with the physician who prescribed it, not a decision to make from an ebook.

A note on fertility. Many people reading this because of a PCOS diagnosis, or a partner's diagnosis, are also trying to conceive, and that makes this subject emotionally loaded in a way that deserves honesty up front. This guide is not fertility treatment. Nothing here promises a pregnancy outcome. Cycle regularity and ovulation are discussed in these pages as physiology, information about how your body is functioning, because that is genuinely useful to understand. If conceiving is your goal, that conversation belongs with your physician or a reproductive endocrinologist, who can evaluate your specific situation properly. Please keep taking any prescribed medication exactly as directed, and talk to your doctor or a qualified practitioner before starting any new supplement, especially if you are pregnant, breastfeeding, or take other prescriptions.

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The Labs Came Back Normal. You Still Do Not Feel Like Yourself.

Somewhere between a normal TSH, a normal fasting glucose, and a doctor's visit that lasted twelve minutes, a lot of people stop asking questions about their hormones. The cycle is irregular but "not a big deal." The energy is gone but "that's just stress." The weight will not move no matter what changes. The libido quietly disappeared sometime in the last few years. For men, the drive and the muscle and the morning energy that used to be there without effort are just gone, and everyone around them says that is what happens after thirty-five.

None of that is a diagnosis. All of it is information.

Hormones are not the root cause of how you feel. They are the messengers. Estrogen, progesterone, testosterone, cortisol, and DHEA are downstream of what is actually happening in your body, including how your cells respond to insulin, how loud your stress response has been running, how efficiently your liver and gut clear used hormones out of circulation, how well you sleep, and where your nutrient status actually stands. Treating a hormone number in isolation, without asking what is driving it, is how people end up on a prescription that manages a symptom without ever touching the mechanism underneath it. That is a case for looking one level deeper, not a criticism of the prescription.

This guide covers three of the most common patterns I see walk through the door. They are polycystic ovary syndrome (PCOS), low testosterone in men under fifty, and the HPA axis, the stress system, and where it intersects sex hormones. It also covers what a hormone test can and cannot actually tell you, because that matters more than most people realize.

Below are eight keys. Each one explains a specific mechanism, in plain language, and what you can do about it at an educational level. The goal is to make sure that the next conversation you have with a practitioner, whether that is my team or your own physician, is a more informed one, rather than to hand you a diagnosis.

Your Hormones Are Messengers, Not the Root Cause

The question almost never asked. Most hormone conversations start and end with a single number, whether that is your testosterone, your estrogen, or your TSH. If the number is in range, the conversation is over. If it is out of range, the fix is usually to replace or suppress that one hormone directly. What almost never gets asked is the more useful question: what is upstream of this number, and is it likely to keep happening again unless that upstream driver changes.

What actually drives the number. Every steroid hormone your body makes, cortisol, DHEA, testosterone, estrogen, and progesterone, starts from the same raw material: cholesterol, converted step by step through a shared enzyme cascade. That cascade does not run in a vacuum. It is shaped by how much inflammation your body is managing, how insulin-sensitive your tissues are, how well your liver clears hormones after they have done their job, what your gut microbiome is doing to hormones in circulation, and how loudly your nervous system perceives ongoing threat or stress. Change any one of those terrain factors and the downstream hormone picture moves, often without anyone touching the hormone itself. This is exactly why two people with an identical low testosterone reading, or an identical PCOS diagnosis, can have completely different underlying drivers, and why a plan that only targets the hormone number tends to help for a while and then plateau.

Where this guide goes from here. The rest of this guide walks through the terrain factors that most commonly drive the three patterns above, which are insulin resistance, chronic stress and the HPA axis, and estrogen and androgen metabolism through the liver and gut. None of it replaces a hormone panel. All of it is meant to make that panel make more sense.

What you are buying here, stated plainly. The nutrients discussed in this guide sit at different evidence tiers, and I grade each one where it appears rather than presenting them as a single uniform block. Ashwagandha has randomized trial evidence behind it. Inositol has trial evidence that the guideline reviewers themselves rated limited and inconclusive. DIM has one long randomized trial measuring a metabolite ratio rather than a symptom outcome. Most of the rest is laboratory work and reasoned clinical practice, which is what happens to compounds nobody can patent: the large outcome trials never get funded, so absence of evidence here is usually absence of a sponsor rather than absence of an effect. Two things follow, and both are worth acting on. Put your money into the terrain work first, sleep, resistance training, blood sugar, and the deficiencies you can actually measure, because that is where the evidence is strongest and it is also the cheapest part of the plan. Then hold anything you add on top of that to a defined endpoint rather than running it indefinitely.

Insulin Resistance Is the Engine Behind Most PCOS

The part the ovary framing misses. PCOS is usually described to patients as an ovarian problem, meaning cysts on the ovaries, irregular periods, and acne or excess hair growth. That description misses the driver behind most of it. PCOS affects roughly 12 percent of women worldwide by the most common diagnostic standard, the Rotterdam criteria, making it one of the most common hormonal conditions in medicine, and it is closely tied to metabolic function, not just the ovaries.¹ A comprehensive 2024 review described PCOS as marked fundamentally by insulin resistance and hyperandrogenism together, a combination that meaningfully raises the risk of early-onset type 2 diabetes and cardiovascular disease over time, alongside the reproductive symptoms most people associate with the diagnosis.²

How insulin drives the pattern. Insulin does two things that directly drive the PCOS pattern. First, at high circulating levels, insulin acts on the ovary's theca cells and amplifies the effect of luteinizing hormone (LH), pushing those cells to produce more androgens (testosterone and its relatives) than they otherwise would. Second, insulin suppresses the liver's production of sex hormone binding globulin (SHBG), the protein that binds up circulating testosterone and keeps it inactive. Less SHBG means more free, active androgen reaching tissue, even without the ovary making more of it. Put those two effects together and you get the hallmark PCOS picture of irregular or absent ovulation, acne, excess facial or body hair growth, and, in many patients, difficulty losing weight, all downstream of insulin resistance rather than a primary ovarian defect.

The highest-leverage work. Because insulin resistance is the driver, addressing it is where the highest-leverage work happens: resistance training and regular movement (both directly improve cellular insulin sensitivity), reducing refined carbohydrate and added sugar intake, adequate protein and fiber at meals to blunt glucose swings, and correcting the nutrient deficiencies (magnesium and vitamin D among them) that commonly track alongside insulin resistance. Myo-inositol and D-chiro-inositol, nutrients naturally present in the body that support insulin signaling, are commonly discussed for PCOS, and it is worth being honest about what the evidence actually shows rather than what the marketing claims. A 2024 systematic review conducted specifically to inform the international PCOS treatment guidelines found real benefit for some metabolic measures and a potential benefit from D-chiro-inositol for ovulation, but rated the overall evidence limited and inconclusive, not the slam-dunk result inositol is often sold as.³ If you have been prescribed metformin for insulin resistance or PCOS, keep taking it exactly as prescribed. The strategies above are additive to metformin, not a substitute for it, and any change to that prescription belongs with the physician who prescribed it.

The HPA Axis and Your Sex Hormones Under Chronic Stress

A term that isn't a diagnosis. “Adrenal fatigue” is one of the most-searched terms in functional health, and it is not a recognized medical diagnosis. A 2016 systematic review that specifically set out to test whether “adrenal fatigue” was a real, substantiated condition reviewed 58 qualifying studies and concluded there is no consistent evidence for it as an actual medical entity, in large part because the cortisol-testing methods used to diagnose it produce conflicting, inconsistent results across the literature.⁴ That does not mean chronic stress has no effect on your hormones. It means the popular framing overstates a real, more nuanced mechanism, and the honest name for what is actually happening is HPA axis dysregulation, a disruption in the communication loop between your brain and your stress hormone system, not a pair of adrenal glands that have run out of gas.

What's actually happening between the two systems. Your hypothalamic-pituitary-adrenal (HPA) axis, the stress system, and your hypothalamic-pituitary-gonadal (HPG) axis, the reproductive hormone system, share upstream signaling and compete for some of the same downstream resources. Under sustained psychological stress, disordered eating patterns, or excessive exercise without adequate recovery, the brain can disrupt the pulsatile release of gonadotropin-releasing hormone (GnRH), which in turn suppresses the downstream signals that drive ovulation. This is well documented in a condition called functional hypothalamic amenorrhea, where cycles stop or become irregular under sustained physical or psychological stress load, with real downstream effects on bone health, mood, and cardiovascular risk if it continues unaddressed.⁵ A related idea you may have heard is “pregnenolone steal,” the theory that chronic stress shunts the shared hormone precursor pregnenolone toward cortisol production at the expense of DHEA and sex hormones. It is a biochemically plausible model, and it is worth being honest that direct human evidence for this specific substrate competition is limited. What is better supported is the broader pattern: the ratio between cortisol and DHEA, sometimes called the cortisol-to-DHEA ratio, is a recognized marker of how your stress system is functioning, and a ratio that skews toward more cortisol and less DHEA is associated with worse outcomes across aging, pain sensitivity, and frailty.⁶ Chronic stress also directly suppresses testosterone production in men, independent of any effect on cortisol.

What actually helps this pattern. Sleep, nervous system regulation, and addressing the actual sources of chronic stress do more for this pattern than any single supplement. Among the more studied adaptogenic herbs, ashwagandha has real, recent evidence behind it: a 2026 meta-analysis of 23 randomized controlled trials found it meaningfully lowered cortisol and, specifically in men, raised testosterone, with no comparable testosterone effect seen in women.⁷ That sex-specific difference matters and is exactly the kind of nuance worth knowing before assuming a supplement works the same way for everyone. This chapter intentionally stays brief on cortisol itself. The full mechanism of chronic stress, cortisol, and energy production is a subject large enough to deserve its own treatment, and I have kept it out of this one deliberately. What matters here is

the intersection: chronic, unaddressed stress is a legitimate upstream driver of cycle irregularity in women and lower testosterone in men, deserving real attention, without needing to invoke a diagnosis that does not exist to take it seriously.

Estrogen Metabolism and Detoxification, and the Estrobolome

The word almost everyone misuses. Estrogen dominance is a term you have probably heard, usually attached to symptoms like heavy or painful periods, breast tenderness, mood swings, and stubborn weight around the hips and thighs. What gets left out of most explanations is that estrogen dominance is rarely about making too much estrogen. It is much more often about clearing it too slowly, and that clearance happens in two places most people never think about: the liver and the gut.

How clearance actually works. After estrogen has done its job, your liver processes it in two phases. Phase 1 hydroxylates the estrogen molecule down one of several pathways, most notably the 2-hydroxy pathway (generally considered the more favorable route) versus the 16-alpha-hydroxy pathway (a more potent, longer-acting route). Phase 2 then attaches a molecule (methylation, via the COMT enzyme, or glucuronidation) that tags the processed estrogen for elimination. Once tagged for elimination, it travels to the gut to be excreted. Here is the part almost nobody explains: certain gut bacteria produce an enzyme called beta-glucuronidase, which can strip that elimination tag back off, effectively re-releasing already-processed estrogen back into circulation instead of letting it leave the body. This gut-microbiome effect on estrogen has a name in the research literature, the estrobolome, and a disrupted, less diverse gut microbiome with elevated beta-glucuronidase activity is a documented contributor to estrogen-related symptoms and disease risk.⁸ Xenoestrogens, estrogen-mimicking compounds from plastics (BPA), certain personal care products (parabens), and some pesticides, add to the total estrogenic load your liver and gut have to clear, which is why reducing exposure to them is a legitimate, low-risk first step regardless of what your labs show.

Where the leverage actually is. Supporting healthy Phase 1 to Phase 2 liver clearance and a healthy gut microbiome are the two levers that matter most here, alongside reducing xenoestrogen exposure where it is easy to do so. Cruciferous vegetables (broccoli, cauliflower, Brussels sprouts) contain compounds that support the more favorable 2-hydroxy estrogen pathway, and DIM (diindolylmethane), a concentrated compound derived from those same vegetables, has real human trial evidence behind it: a randomized, placebo-controlled trial found that daily DIM shifted the urinary estrogen metabolite ratio toward the more favorable pathway and raised SHBG, compared to placebo, over twelve months.⁹ That same trial is also a genuine safety reminder rather than just a win: it was conducted in women taking tamoxifen, a prescription anti-estrogen medication, and DIM measurably reduced the levels of active tamoxifen metabolites in their blood. If you are on tamoxifen, another prescribed hormone therapy, or any medication that affects estrogen metabolism, that is a conversation for your prescriber before adding DIM or any similar compound, not a decision to make on your own. Calcium-D-glucarate, a compound that inhibits the same beta-glucuronidase enzyme responsible for re-releasing estrogen in the gut, is the other commonly discussed nutrient in this category. As for how much this actually matters for long-term risk, it is worth being precise rather than

alarmist: a case-control study of the 2-hydroxy to 16-alpha-hydroxy estrogen metabolite ratio found a significant association with breast cancer risk in postmenopausal women, but not in premenopausal women, where naturally fluctuating hormone levels across the cycle appear to make the ratio less meaningful as a standalone marker.¹⁰ That finding is genuine, and it is also a narrower one than the way this topic often gets marketed.

A note on pregnancy and breastfeeding. Several of the compounds discussed in this guide, including DIM, calcium-D-glucarate, berberine, and ashwagandha, are not recommended during pregnancy or while breastfeeding. If you are pregnant, trying to conceive, or nursing, talk to your physician before starting any of them.

Androgens Are Not Just “The Male Hormones”

Two blind spots from one assumption. Testosterone gets treated culturally as the male hormone and estrogen as the female one, which leads to two blind spots: women’s androgen levels get almost no clinical attention unless they are too high (as in PCOS), and men’s estrogen, produced in meaningful amounts via a process called aromatization, gets almost none at all. Both sexes make, need, and are affected by the full spectrum of sex hormones.

What androgens actually do for each sex. Androgens, testosterone, DHEA, androstenedione, and DHT among them, play a real role in women’s health beyond reproduction. Bone density and remodeling, cardiovascular health, muscle mass and tone, and cognitive function all depend in part on adequate androgen levels, not just adequate estrogen.¹¹ Low androgen states in women, whether from surgical removal of the ovaries, adrenal insufficiency, or the natural decline in DHEA-S with age, are a real and underappreciated pattern, distinct from the high-androgen picture seen in PCOS. In men, the enzyme aromatase converts a portion of circulating testosterone into estradiol, and adipose (fat) tissue is a major site of that conversion, meaning body composition directly affects a man’s estrogen-to-testosterone balance, not just his testosterone number in isolation.

What this means for you, by sex. For women, the practical takeaway is that a “normal” estrogen and progesterone panel does not tell the whole story, and androgen status (total and free testosterone, DHEA-S) is worth including in a complete hormone workup, particularly if energy, libido, or muscle mass are part of the symptom picture. For men, body composition, addressed in the next key, is directly relevant to androgen balance in both directions, not just the testosterone number. Any decision to supplement DHEA or pursue hormone therapy of any kind, for either sex, is a conversation with a practitioner working from your actual labs, not a decision to make from general information in a guide like this one.

What Is Actually Driving Low Testosterone in Men Under Fifty

The demographic shift nobody predicted. Low testosterone used to be framed as an older man's condition. It increasingly is not. A population-level study tracking American men across multiple survey years found that testosterone levels have declined across birth cohorts independent of age itself, a decline not explained by the usual suspects like rising obesity or smoking rates.¹² Something in modern life, beyond simple aging, is lowering testosterone in younger men, and the good news is that several of the biggest drivers are genuinely modifiable.

The biggest modifiable levers. Sleep is the most underrated lever here. In a controlled study, healthy young men restricted to five hours of sleep per night for one week saw daytime testosterone levels drop by 10 to 15 percent, a decline larger than the 1 to 2 percent per year testosterone naturally falls with normal aging.¹³ Body composition is the second major lever: obesity is the clinical condition most strongly associated with lowered testosterone in men, working through two separate mechanisms depending on severity, reduced SHBG in modest excess weight, and genuine suppression of the brain-testicular signaling axis in more significant obesity, driven partly by inflammatory signaling and partly by the aromatase conversion of testosterone to estrogen in fat tissue described in the previous key.¹⁴ Blood sugar dysregulation compounds both of the above, since insulin resistance independently suppresses SHBG in men, the same mechanism that raises free androgens in women with PCOS, just running in the same biochemical direction toward a different clinical picture. Chronic stress adds a fourth layer, suppressing testosterone production through the same HPA-HPG crosstalk covered in Key 3. Nutrient status matters too, though it is worth being precise about which nutrients actually have evidence behind them and which are oversold. A 2025 study of young men found several trace elements, including zinc, iron, and sulfur, associated with more favorable testosterone levels, while others, including phosphorus and strontium, were associated with lower levels.¹⁵ Vitamin D is a more honest story than most wellness content admits: the most rigorous recent meta-analysis of randomized controlled trials, published in 2026, found no clear effect of vitamin D supplementation on total testosterone or related markers in adult men, despite how often vitamin D is marketed as a testosterone booster.¹⁶ Correcting a genuine vitamin D deficiency still matters for bone, immune, and overall metabolic health. It is just not, on the current evidence, a reliable lever for testosterone specifically.

Where to focus first. In order of leverage: prioritize seven to nine hours of consistent sleep, build and maintain lean muscle through resistance training while addressing excess body fat, stabilize blood sugar through the same strategies covered in Key 2, and manage chronic stress directly rather than assuming it will resolve on its own. Ashwagandha, covered in Key 3, is one of the few supplements with real randomized-trial evidence for raising testosterone specifically in men.⁷ If you are already on prescribed testosterone replacement therapy (TRT), or your physician has raised it as an option, nothing in this guide is a reason to start, stop, or adjust that treatment yourself. Low testosterone that is confirmed on repeat morning

bloodwork, alongside real symptoms, is a legitimate medical finding that deserves a proper workup with your physician, not a self-diagnosis from a symptom list, and whether testosterone therapy is appropriate for your specific case is that physician's call to make with you.

Thyroid, the Cross-Referenced Driver

Where thyroid and sex hormones overlap. Thyroid function and sex hormone function are wired together more closely than most people realize, which means an unaddressed thyroid issue can look, on the surface, exactly like a primary sex hormone problem.

How the two systems are wired together. Thyroid hormone status influences the menstrual cycle both directly, through effects on the ovary, and indirectly, through its effect on sex hormone binding globulin (SHBG), prolactin, and gonadotropin-releasing hormone (GnRH) signaling.¹⁷ That means an underactive or autoimmune thyroid can drive irregular cycles, low libido, and fatigue that look identical to a primary hormonal pattern, and a hormone panel alone will not distinguish between the two. Hashimoto's thyroiditis, an autoimmune condition and the most common cause of hypothyroidism, is common enough among people already dealing with PCOS and other hormonal patterns that it is always worth ruling in or out, not assumed away because a single TSH came back "normal."

Check before you assume. If thyroid symptoms (cold intolerance, hair thinning, unexplained weight change, constipation, or persistent fatigue despite a full sleep and stress correction) are part of your picture, a full thyroid panel, not just TSH, is worth pursuing before assuming the pattern is purely a sex hormone issue. The complete framework for autoimmune thyroid disease, its triggers, its testing, and its treatment sequence lives in the companion guide, *Unlock Your Hashimoto's*. This guide will not re-cover that ground. The short version relevant here is simply this: do not let a "normal" TSH close the door on thyroid as a contributing driver of your hormonal symptoms.

Testing Honestly, and What Each Sample Type Can and Cannot Tell You

The differences that actually matter. Not all hormone tests measure the same thing, and the differences matter more than most patients are told. Choosing the wrong sample type for the question you are actually asking is one of the most common, and most avoidable, sources of confusing or contradictory hormone results.

What each sample type actually shows. Serum (blood) testing measures total hormone levels, bound and unbound, at a single point in time. It is the current standard of care for monitoring hormone replacement therapy and is the only sample type that reliably captures SHBG alongside your sex hormones, but a single blood draw cannot show you a cortisol rhythm across the day, and it misses most free, unbound hormone and metabolite information entirely. Urine testing captures hormone metabolites, meaning it can show you how your body is actually processing hormones (which liver pathway your estrogen is taking, for example) as well as how much hormone is circulating. Interpretation requires care. A low result can reflect either low production or increased tissue uptake, and a high result can point to overproduction or to a backup in clearance. Saliva testing reflects the free, bioavailable fraction of a hormone rather than the bound majority, which makes it genuinely useful for tracking a cortisol rhythm across the day, but it does not measure hormone metabolites, and, as the “adrenal fatigue” research above makes clear, the cortisol-testing literature built on salivary methods has produced inconsistent, sometimes conflicting results across studies, which is a real limitation worth naming rather than glossing over.⁴ None of these three approaches is universally “better.” Each answers a different question, and the right choice depends on what you are actually trying to find out.

Matching the test to the question. If you are being screened for a first-pass hormonal workup, serum testing covering your core sex hormones, SHBG, and a full thyroid panel is a reasonable, well-validated starting point. If cycle-related patterns or estrogen metabolism specifically are the question, a urine-based hormone metabolite panel adds real information a single blood draw cannot. If cortisol rhythm across the day is the specific question, salivary or urine testing done properly, at the right times of day, across a full cycle if you are still menstruating, adds value that a single morning blood cortisol cannot. What no single test does, regardless of sample type, is diagnose you on its own. Testing informs a clinical picture. It does not replace one.

Red Flags: When to Seek Care Now, Not Later

This guide is educational and does not diagnose. Some symptoms warrant a prompt conventional medical evaluation rather than a supplement or a wait-and-see approach, regardless of what else is going on hormonally:

- **Postmenopausal bleeding** (any vaginal bleeding after menopause) always warrants prompt evaluation.
- **A rapidly virilizing presentation** (fast-developing deepening of the voice, significant new hair growth, or clitoral enlargement over weeks to months, rather than a gradual pattern over years) can signal an androgen-secreting tumor and needs prompt workup.
- **Severe, acute pelvic pain**, especially sudden in onset, needs same-day evaluation.
- **A new testicular mass or lump** always warrants prompt evaluation and should never be watched and waited on.
- **Galactorrhea** (spontaneous milk-like nipple discharge unrelated to pregnancy or nursing) can indicate a pituitary issue and warrants a prolactin level and follow-up.
- **Adrenal insufficiency (Addison's disease) and pituitary disease** are real, distinct medical conditions that can present with fatigue and low hormone levels that look similar to the patterns in this guide. They require conventional diagnostic workup, and this guide is not a substitute for ruling them out if your presentation or labs suggest either one.

None of the above are things to address with supplements or lifestyle change first. See a physician promptly.

The Staged Roadmap

Hormonal patterns do not resolve on a fixed calendar, and healing is rarely a straight line. The general pace below reflects realistic timelines for the mechanisms covered in this guide, not a guarantee for any individual case.

Weeks 1 to 4: Foundation. Sleep consistency, blood sugar stabilization (protein and fiber at meals, reduced refined carbohydrate), reducing xenoestrogen exposure, and beginning to address the biggest single stress load in your life. Early energy and sleep improvements are typically the first thing people notice in this window.

Months 2 to 3: Cycle and pattern changes. For women with PCOS-driven irregularity, this is typically when cycle-length changes become noticeable, often across two to three cycles, as insulin sensitivity begins to shift. For men, this is when body composition changes from consistent resistance training and sleep begin to show up in energy and libido.

Months 3 to 6: Metabolic and hormonal consolidation. This is the window where insulin sensitivity, estrogen clearance patterns, and cortisol rhythm changes typically stabilize into a new, more consistent baseline, assuming the foundational work from the first two phases has been consistent.

Months 6 to 12: Full pattern stability. Full hormonal rebalancing, for both the PCOS and low-testosterone patterns covered here, typically takes six to twelve months of consistent terrain work to reach a stable, durable baseline, longer in more severe or longstanding cases. This is also the point where repeat lab testing is most useful for confirming what your body has already started telling you symptomatically.

If you are not seeing meaningful movement by the three-month mark, that is useful information too. It usually means an upstream driver has not yet been identified, not that the approach has failed.

Cycle and Symptom Tracking Worksheet

Whether you are tracking a menstrual cycle or simply tracking symptoms day to day, consistent tracking is one of the most useful, lowest-cost tools available, both for your own awareness and for a more productive conversation with any practitioner you see. Copy this table (paper, spreadsheet, or notes app) and fill it in daily.

Date	Cycle day (or N/A)	Sleep (hours, quality 1 to 5)	Energy (1 to 5)	Mood (1 to 5)	Stress (1 to 5)	Libido (1 to 5)	Skin, hair, or weight changes	Other symptoms	Notes

Monthly reflection prompts

1. Any change in cycle length, regularity, or flow this month compared to last month (if applicable)
2. Your average sleep quality, and anything specific that improved or disrupted it
3. The single largest stress load this month, and how you handled it
4. Whether energy and mood tracked together, or one lagged behind the other
5. What changed in your routine this month (diet, training, sleep, supplements, medications) that might explain any shifts above

Bring this log to your next Root Discovery Session or physician visit. A month of consistent tracking tells a practitioner more than a single lab draw ever can on its own.

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a free consultation to review your specific situation: your symptom history, what testing (if any) you have already had, and what a reasonable next step looks like for your case. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Starting on your own

If you would rather start on your own, start by tracking. The cycle and symptom tracking worksheet earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you what a single lab draw cannot: how symptoms move across a cycle or across a month, which is exactly the context that makes hormone testing interpretable instead of a snapshot. Two weeks of honest data makes any future appointment, with our team or your own physician, dramatically more useful than memory alone.

Start your free tracker at rootcauseunlocked.com/tracker.

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This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment, and it is not fertility treatment. Always consult your physician before starting any new supplement or exercise program, particularly if you are pregnant, breastfeeding, or taking prescription medications, including hormonal birth control, testosterone replacement therapy, or metformin. Continue any prescribed medication exactly as directed by your physician.

References

Every citation below was independently retrieved and confirmed against its live PubMed record (eSearch plus eSummary and/or eFetch, including full-text verification via PubMed Central where the indexed abstract was incomplete) before use. Full verification detail, including claims investigated and left uncited, is in the companion file Hormones-Citation-Ledger.md.

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For educational purposes only. This guide does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship.

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From Understanding to Doing

Everything up to this point was about understanding what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing in this kit is a reason to stop or change a prescription. Not birth control, not testosterone replacement, not metformin, not thyroid medication. If you want to change any of those, that is a conversation with the prescriber who started them.

This is not fertility treatment. If you are trying to conceive without success, that calls for a reproductive endocrinologist, and time matters. Do not spend a year on supplements first.

See a doctor promptly for: a breast lump, unexplained vaginal bleeding after menopause or between periods, bleeding after intercourse, sudden severe pelvic pain, visual changes with headaches, unexplained weight loss, or new severe fatigue with dizziness and darkening of the skin. That last combination can indicate adrenal insufficiency, which is a genuine and serious diagnosis.

The one idea this whole kit rests on

Key 1 of *Unlock Your Hormones* says it: **hormones are messengers, not the root cause.**

A hormone panel tells you what the messengers are doing. It rarely tells you why. Testosterone that is low because of poor sleep, excess weight, and untreated apnea reads the same on paper as a pituitary-driven low, and those need completely different responses. Estrogen that looks unbalanced because of insulin resistance is not the same problem as an imbalance with a structural cause.

So the useful question is never “what is my hormone level.” It is “what is making it do that.”

The eight keys, and what tests each one

Key	What it means	How you find it
1. Messengers	Hormones report on the system, they are not the system	The framing for everything below
2. Insulin	Insulin resistance is the engine behind most PCOS	Fasting insulin, glucose, HbA1c. The highest-yield panel here
3. HPA axis	Chronic stress and its effect on sex hormones	Contested. See the honest section below
4. Estrogen metabolism	Clearance, the gut, and the estrobolome	Metabolite panels, read as routes rather than targets
5. Androgens	Not just a male concern	Total and free testosterone, SHBG, DHEA-S
6. Low testosterone in men	What is actually driving it under fifty	Morning total testosterone, repeated , then a cause hunt
7. Thyroid	The cross-referenced driver	TSH, free T4, free T3, antibodies
8. Testing honestly	Serum versus dried urine, and what each is good for	The section below is the one to read first

Key 8: the honest read on serum, saliva and urine tests

I am putting this first because it decides how much money you spend on everything else, and because the honest answer is more interesting than either side usually admits.

Serum testing is what most clinical decisions are made on, what most reference ranges are built from, and what your physician will act on. It remains the default.

Dried urine panels are widely used in functional medicine, and the common criticism of them, that they are simply unvalidated, is not accurate. Here is what the published record actually shows.

What the assay validation shows, which is real

A method study compared dried urine analyzed by gas chromatography with tandem mass spectrometry against serum, and against 24-hour collection. It reported good reliability between the dried urine and serum assays, urine metabolite concentrations following the same pattern as serum across the menstrual cycle in a subset of women, and intraclass correlation coefficients above 0.95 comparing the four-spot method with 24-hour collection and dried with liquid urine, which is excellent agreement. Its conclusion was that for estradiol and progesterone the dried urine assay is a good surrogate for serum testing (PMID 31384769).

A second study extended this across a wider panel, finding dried versus liquid urine agreement above 0.90 for reproductive hormones and 0.75 to 0.99 for organic acids, and four-spot versus 24-hour agreement above 0.9 for 14 of 17 metabolites with no systematic differences between collection methods (PMID 33722278).

A third study showed that dose-proportional changes in estrogen exposure from transdermal estradiol gel could be seen in the urinary estrogen profile, and distinguished the profiles of premenopausal women, postmenopausal women on estrogen therapy, and postmenopausal women on none (PMID 35483542).

So the assay measures what it claims to measure, reliably, and a four-spot dried collection can stand in for a 24-hour jug. That is genuine analytical validation and it deserves to be stated plainly.

What that does not establish, which is the part that matters for you

Analytical validation and clinical validation are different things.

Analytical validation asks whether the assay accurately and reproducibly measures the molecule. The answer above is yes.

Clinical validation asks whether making treatment decisions from those results leads to better outcomes than not doing so.

None of those studies answer that question, and I could not find any that answer it. The point is about what a measurement can justify on its own, not a criticism of the assay.

Two more things you should hear from me rather than discover later.

Who did the research. All three studies were conducted by the laboratory that sells the test, and the papers disclose it: the founder and owner is an author on each. Two include Frank Stanczyk, an independent steroid assay researcher whose involvement is meaningful, and one notes that the statistics were calculated and interpreted by an independent agent. Those disclosures are to their credit. It also remains true that company-run validation of a company's own product is weaker than independent replication, and independent replication is what I could not find.

How the reference ranges are built, which is the real caveat. This is not my inference. The manufacturer's own interpretive guide states it. The ranges come from their internal database with their own exclusions applied, and the guide says the ranges are deliberately **set more tightly than classic diagnostic ranges so as to be optimally used in functional medicine practice**, giving the example that the upper limit for testosterone is set at the 80th percentile in women and the 90th in men.

Read that carefully, because it is the crux. A classic diagnostic range is built to separate disease from health. A percentile cutoff chosen for clinical utility is built to flag people for attention. **Being outside such a range means you are in a minority of the tested population. It does not mean you have a disorder.** The manufacturer is transparent about this, and it is the single most useful thing to understand before you read your own report.

What I would actually do with one

- **If a decision matters, anchor it on serum**, particularly anything involving a diagnosis or a prescription.
- **Use dried urine for what the validation actually supports**, such as tracking a pattern over time, monitoring estrogen exposure on therapy, and seeing metabolite routes that serum does not show.
- **Treat an out-of-range metabolite as a prompt to look closer, not as a diagnosis**, for the reference-range reason above.
- **Do not repeat it often**. It is most informative as a baseline and a follow-up, not as a monthly scorecard.

The one to still be careful about: cortisol panels sold to diagnose “adrenal fatigue”

This is a separate claim from everything above, and the validation does not transfer to it.

The assay can measure cortisol and its metabolites accurately. That does not make **adrenal fatigue**, meaning stress-exhausted adrenals that under-produce, a recognized clinical diagnosis. It is not one. A pattern on a cortisol curve is not a disease, and measuring something accurately does not validate the interpretation sold on top of it.

Adrenal insufficiency is a real and serious diagnosis, established with morning cortisol and specialist testing rather than a wellness panel. If you have severe fatigue with dizziness, low blood pressure, salt craving, and darkening skin, that is a physician now, not a kit.

Key 2: insulin, the engine behind most PCOS

If you have PCOS, this is the highest-yield section in this guide.

What to ask for. Fasting insulin, fasting glucose, and HbA1c together. Fasting glucose alone will look normal for years while insulin climbs to keep it there, which is exactly the window in which something can be done.

Why it matters more than the sex hormones. The international guideline framing puts lifestyle intervention as first-line therapy in PCOS, and a modest weight loss of 5 to 10% can significantly reduce reproductive, metabolic, and psychological symptoms, while the same source is candid that achieving it remains a challenge for patients and clinicians alike (PMID 32814159).

Read both halves of that. The target is smaller than people assume, and the difficulty is genuine and acknowledged, not a failure of willpower.

PCOS diagnosis, briefly

PCOS is diagnosed on a combination of irregular ovulation, clinical or biochemical androgen excess, and ovarian appearance on ultrasound, with other causes excluded. **You do not need all three.** Two of three, with other causes ruled out, is the standard framework, and the exclusions matter, because thyroid disease, elevated prolactin, and non-classical congenital adrenal hyperplasia can all imitate it.

What to ask for alongside the insulin panel. Total and free testosterone, SHBG, DHEA-S, LH and FSH, prolactin, TSH with free T4, and 17-hydroxyprogesterone.

Key 6: low testosterone in men, and the diagnostic discipline that is usually skipped

How common it is. Prevalence estimates suggest 6 to 12% of men aged 40 to 49 have biochemically low testosterone, commonly associated with obesity, metabolic syndrome, and sleep apnea (PMID 41127744).

Read that association list, because this key rests on it. In men under fifty, low testosterone is far more often a consequence of something than a primary gland problem. Excess weight, poor sleep, untreated sleep apnea, and metabolic syndrome all suppress it, and all of them are addressable.

The diagnostic discipline. The Endocrine Society clinical practice guideline on testosterone therapy in men with hypogonadism sets out how diagnosis and monitoring should be done, including monitoring that evaluates symptoms, adverse effects and compliance, measures serum testosterone and haematocrit, and evaluates prostate cancer risk in the first year of therapy (PMID 29562364).

The two rules most often broken:

1. **Draw it in the morning.** Testosterone has a daily rhythm and an afternoon sample can be misleadingly low.
2. **Confirm it on a second, separate morning sample** before anyone acts on it. A single low reading is not a diagnosis.

And if it is genuinely low, find the cause before deciding on a response. That means LH and FSH to separate a testicular cause from a pituitary one, prolactin, ferritin and iron studies, and a sleep apnea assessment. Starting replacement without that cause hunt can bury a finding that mattered.

If you are already on testosterone replacement, nothing in this kit is a reason to stop it. That is a prescriber conversation, and stopping abruptly has consequences.

Key 3: the HPA axis, without the marketing

Chronic stress genuinely interacts with sex hormones. That is established physiology, and it is Key 3 of the ebook.

What does not follow from it is that a saliva panel can measure your “adrenal fatigue” and a supplement can fix it. Covered above in Key 8.

What is worth doing instead. Treat load as a clinical variable. Sleep, work hours, caregiving, training volume, and undereating are all inputs, and all of them are more tractable than a cortisol curve. If there is genuine clinical suspicion of adrenal insufficiency, that is a morning cortisol and a physician, not a wellness panel.

Keys 4, 5 and 7: the rest of the panel

Estrogen metabolism and the estrobolome (Key 4). The gut participates in estrogen recirculation, which is established biology and the rationale for the fiber emphasis in the diet guide. On testing: the metabolite measurements themselves are analytically sound, as covered in Key 8, and what has not been established is that steering treatment by metabolite ratios changes outcomes. So read the ratios as information about routes rather than as targets to correct. Fiber, bowel regularity, and alcohol moderation are defensible on their own terms without needing a ratio to justify them.

Androgens in women (Key 5). Total and free testosterone, SHBG, and DHEA-S. SHBG matters more than people expect, because it determines how much of your testosterone is actually free and active, and it falls with insulin resistance. A normal total testosterone with a low SHBG can mean a high free testosterone, which is the pattern that explains symptoms.

Thyroid (Key 7). TSH with free T4 and free T3, plus TPO and thyroglobulin antibodies if never checked. Thyroid disease imitates and worsens most of what is in this kit. As in the other kits, I have deliberately not published functional target ranges for thyroid markers, because the right target depends on age, pregnancy, and existing replacement.

Two markers in that table that get read wrong

Homocysteine, read correctly. Homocysteine is a methylation-cycle marker. It rises when B12, folate, or B6 run short, and with thyroid or kidney trouble. Do not let anyone wave off an elevated result as trivia. In my read this is one of the most consequential numbers in the panel, because it tells you the methylation machinery is running short somewhere findable. The way to act on it is not to chase the number with blind supplementation. Find which input is short, B12, folate, B6, thyroid, or kidney function, correct that, and let homocysteine confirm the fix on retest. My functional target of 5 to 7 in the table below reflects how seriously I take it. It is a window, not a floor. Lower is not better: a homocysteine driven below that window points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result.

hs-CRP, read for exactly what it measures. It is a nonspecific screening marker. An elevated result tells you an inflammatory load exists somewhere. It does not tell you where it is coming from, and it never confirms that your hormones are the source. It is worth having because a load you did not know about changes what you look for next, not because the number itself points anywhere.

Functional ranges for the metabolic markers

Clinical targets I use at Root Health, not validated diagnostic thresholds. Sex hormone reference ranges are deliberately not listed here, because they depend on sex, age, cycle day, and menopausal status, and a single table would mislead more than it helps. Your lab's ranges and your clinician's reading are what count for those.

Marker	Units	Functional target	Approximate standard range
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0
Ferritin	ng/mL	50 to 150	25 to 300
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Vitamin B12	pg/mL	500 to 900	200 to 1100
Homocysteine	umol/L	5 to 7	under 10.0
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Zinc (plasma)	ug/dL	70 to 110	60 to 120
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

Reading your results as a pattern

Pattern one: PCOS with insulin resistance. Irregular cycles, androgen excess, fasting insulin creeping up, SHBG low. This is the most common and most treatable pattern in the kit, and the diet and supplement guides are built around it.

Pattern two: low testosterone with a cause attached. A man under fifty with low morning testosterone, excess central weight, and snoring. The testosterone is downstream. Treat the apnea and the weight and re-measure before deciding about replacement.

Pattern three: normal panel, real symptoms. Common. It moves the work to sleep, load, blood sugar stability, and thyroid, and it is not a dismissal.

Pattern four: something that needs a specialist. Very high testosterone or DHEA-S, high prolactin, visual changes, or postmenopausal bleeding. Those go to a physician, not to a protocol.

The conversation with your doctor

A request list you can bring

If PCOS is the question. Fasting insulin, fasting glucose, HbA1c, total and free testosterone, SHBG, DHEA-S, LH, FSH, prolactin, TSH with free T4, 17-hydroxyprogesterone.

If low testosterone is the question. Morning total testosterone, repeated on a second morning, plus free testosterone, SHBG, LH, FSH, prolactin, ferritin and iron studies, HbA1c, and a sleep apnea assessment.

For everyone. 25-hydroxy vitamin D, B12, hs-CRP, and a CBC.

The sentence that helps

“My fasting glucose is normal. Could we add a fasting insulin, because I understand glucose stays normal for a long time while insulin is compensating?” That is specific, accurate, and it is the test most often left off.

What these labs cannot tell you

- **Why your hormone is doing what it is doing.** That is Key 1, and it is what matters most.
- **Whether you have adrenal fatigue.** Not a recognized diagnosis.
- **Whether a urine metabolite pattern should change your treatment.** The measurement is sound. The outcome evidence for steering treatment by it is what is missing, and Key 8 explains the difference.
- **Whether you can conceive.** That is a fertility evaluation, and it is time-sensitive.

When to seek clinical review

Promptly. Breast lump, postmenopausal bleeding, bleeding between periods or after intercourse, sudden severe pelvic pain, visual changes with headache, unexplained weight loss, or severe fatigue with dizziness, low blood pressure, salt craving and skin darkening.

Soon. No periods for three months outside pregnancy, rapidly worsening hirsutism or male-pattern hair loss, galactorrhoea, or twelve months of trying to conceive without success, six months if you are over 35.

Your next step

The companion guides do the rest: **The Hormone Diet Guide** covers insulin resistance and the fiber and estrobolome work, and **The Hormone Supplement Guide** covers inositol, NAC, and what the evidence does and does not support.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, track your cycle and symptoms together. In this category the pattern across three cycles is worth more than any single panel:

Free symptom tracker: rootcauseunlocked.com/tracker

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Educational content only. Not medical advice, not a diagnosis, not fertility treatment, and not a reason to change any prescription.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing here is a reason to stop or change a prescription, including birth control, testosterone replacement, metformin, or thyroid medication.

This is not fertility treatment. If you are trying to conceive without success, that is a reproductive endocrinologist, at twelve months of trying or six months if you are over 35.

If you are pregnant, nursing, underweight, or have a history of disordered eating, talk with a clinician before changing how you eat. **This guide discusses weight, and it does so carefully, for reasons in the next section.**

See a doctor promptly for a breast lump, postmenopausal bleeding, bleeding between periods, sudden severe pelvic pain, or visual changes with headache.

Before anything else, about weight

The guideline literature puts lifestyle intervention as first-line therapy in PCOS, and a **weight loss of 5 to 10% can significantly reduce reproductive, metabolic, and psychological symptoms**, while the same source states plainly that achieving it remains a challenge for both patients and healthcare providers (PMID 32814159).

Two things follow, and I want both on the record.

The target is smaller than most people are told. Not a transformation. Five to ten percent.

And the difficulty is genuine and documented, not a character flaw. The exercise position statement for PCOS specifically asks clinicians to account for the significant psychological burden, including **weight stigma**, and the high prevalence of comorbidities, when delivering interventions (PMID 38960811).

If you have spent a decade being told to lose weight by people who did not help you do it, that position statement is on your side. This guide tries to be the help rather than the instruction.

And if weight is not your issue, which is true for a meaningful share of people with PCOS, skip this and go to Part Two. Lean PCOS exists, and the insulin work still applies.

Where food sits among the eight keys

Unlock Your Hormones works through eight root causes. Food is a direct lever on Key 2, insulin resistance, which the ebook calls the engine behind most PCOS, and a real contributor to Key 4, estrogen metabolism and the estrobolome. It supports Key 6 in men, where weight and metabolic health drive much of what looks like a testosterone problem.

It is neither a lever on thyroid autoimmunity nor a treatment for infertility. The lab guide maps all eight keys.

Part one: eating for insulin, which is the engine

If you take one thing from this guide, take this part.

The goal is a diet that does not spike insulin repeatedly across the day, rather than a low-carbohydrate diet. Those are different, and the insulin-steady version is much easier to live with.

The five changes that do most of the work:

1. **Protein at every meal, starting with breakfast.** This single change flattens the rest of the day more reliably than anything else here.
2. **Never eat carbohydrate naked.** Pair it with protein, fat, or fiber. Toast alone and toast with eggs behave completely differently.
3. **Fiber at every meal.** Vegetables, legumes, whole grains, and fruit with the skin on. Fiber does double duty here, and Part Three explains the second job.
4. **Cut liquid sugar first.** Sweetened drinks, juice, and the coffee order that is functionally a dessert. This is usually the largest single input and the easiest to remove.
5. **Walk after meals.** Ten to fifteen minutes. It blunts the post-meal glucose rise and it is the least demanding exercise intervention that exists.

What to expect and when. Energy and afternoon crashes often shift within two weeks. Cycle changes take cycles, usually three to six months. Judge accordingly.

Sourcing note. The five practices above are standard clinical nutrition for insulin resistance rather than findings from a PCOS-specific trial, and I have not attached citations to them for that reason. The guideline-level claim they serve, that lifestyle is first-line and 5 to 10% weight loss meaningfully reduces symptoms, is cited above (PMID 32814159).

Part two: movement, because it is a hormonal intervention here

Exercise in PCOS has its own position statement (PMID 38960811), which is unusual and tells you the evidence base is taken seriously.

What matters most, practically:

- **Resistance training two or three times a week.** Muscle is where glucose goes. This is the highest-value form of exercise for insulin resistance and it is chronically under-prescribed to women.
- **Walking daily**, especially after meals.
- **Something you will still be doing in a year.** Adherence beats optimality, and the position statement's point about psychological burden applies here too.

For men with low testosterone, the same logic runs harder. A scoping review of nonpharmacological interventions concludes that proper exercise, sleep, and diet are indisputable approaches on general health grounds, while nutraceuticals rest on weaker evidence and need personalizing (PMID 35810031). Weight, sleep, and apnea are the levers, and they are the ones a capsule cannot reach.

Part three: how fiber and the gut shape estrogen

This is Key 4, where diet does something specific rather than general.

The mechanism, plainly. Estrogen is conjugated in the liver and sent into the bowel for elimination. Gut bacteria can deconjugate some of it, allowing reabsorption. That recirculation loop is the estrobolome, and the two things that most influence how much gets reabsorbed are transit time and fiber.

What follows

- **Fiber, aiming for 25 to 30 g daily**, built up gradually so you do not simply bloat.
- **A daily bowel movement.** Unglamorous, and directly relevant to the loop above.
- **Cruciferous vegetables** several times a week. Broccoli, cabbage, cauliflower, Brussels sprouts.
- **Alcohol moderation**, because the liver is doing the conjugation step.

What I am not going to tell you. That you need a urine estrogen metabolite panel to guide any of this, or that a “detox” supplement corrects the ratios on one. The lab guide covers that testing fairly: the measurements are analytically sound, and what has not been shown is that steering treatment by the ratios changes outcomes. Fiber, regularity, and alcohol moderation are defensible on their own terms and do not need a ratio to justify them.

Part four: a week that actually works

Breakfast, the highest-leverage meal. Eggs with vegetables. Greek yogurt with berries and seeds. Leftover dinner protein. Not cereal, not toast alone, not juice.

Lunch. A protein, a large volume of vegetables, a fiber-rich carbohydrate. Lentil soup with bread. Chicken and bean salad. Leftovers from last night.

Dinner. Protein, two vegetables, a starch you tolerate. Cook enough to cover tomorrow's lunch.

If you snack, pair it. Apple with nut butter, not apple alone.

The batch habit. Cook two proteins and roast a tray of vegetables on one day. It removes the decision at the point when you have the least capacity to make it.

What I deliberately left out, and why

- **Very low carbohydrate and ketogenic diets as the default.** They can help insulin resistance and they are demanding, and the guideline framing is lifestyle intervention with a 5 to 10% weight loss target rather than a specific macronutrient extreme (PMID 32814159). If you want to try one, it is a conversation, not a starting point.
- **Dairy and gluten elimination for everyone with PCOS.** Widely recommended, and I did not find evidence supporting blanket removal for this condition. Test if you have a reason.
- **Estrogen “detox” diets built on metabolite panels.** See Part Three.
- **Seed cycling.** Popular and I could not find trial evidence for it. Eating seeds is fine. The cycling schedule is the part without support.
- **Extreme calorie restriction.** It suppresses the reproductive axis, which is the opposite of the goal, and it makes the 5 to 10% harder to sustain.
- **Anything framed as a fertility diet.** If conception is the goal, the appointment is the intervention.

Telling you what to skip is part of what you paid for.

Running a fair experiment

Cycles 1 and 2, baseline. Change nothing. Track cycle length, symptoms, energy, and the things that bother you most.

Cycles 3 to 6, the insulin work. Protein at breakfast, pair the carbohydrate, fiber up, liquid sugar out, walk after meals.

Cycle 7, judge it. Compare cycles, not days. Cycle regularity is the outcome that matters most and it is slow.

Realistic expectations

What is realistic. Steadier energy in two weeks. More regular cycles over three to six months. Gradual improvement in metabolic markers. Slow improvement in hirsutism, because hair growth is slow.

What is not realistic. Diet curing PCOS, which is a long-term condition that gets managed. Diet fixing testosterone that is low because of untreated sleep apnea. Any of this substituting for a fertility evaluation.

When to seek professional care

Promptly. Breast lump, postmenopausal bleeding, bleeding between periods or after intercourse, sudden severe pelvic pain, visual changes with headache.

Soon. No periods for three months outside pregnancy, rapidly worsening hirsutism or hair loss, milk discharge from the breast, twelve months trying to conceive (six if over 35), or a diet that has narrowed below roughly twenty foods.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your labs, your cycles, your medication list, and whether conception is a goal.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, track cycles and symptoms together. Three cycles of data is worth more than any single panel:

Free symptom tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your Hormone Labs** covers the fasting insulin most often left off the panel, and **The Hormone Supplement Guide** covers inositol and NAC with their honest evidence grades.

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Hormones-Kit-Citation-Ledger.md`.

1. Lie Fong S, Douma A, Verhaeghe J. Implementing the international evidence-based guideline of assessment and management of polycystic ovary syndrome (PCOS): how to achieve weight loss in overweight and obese women with PCOS? *J Gynecol Obstet Hum Reprod*. 2021;50(6):101894. PMID 32814159
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Educational content only. Not medical advice, not a diagnosis, not fertility treatment, and not a reason to change any prescription.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing here is a reason to stop or change a prescription. Not birth control, not testosterone replacement, not metformin, not thyroid medication.

This is not fertility treatment. If you are trying to conceive and it is not happening, see a reproductive endocrinologist. Twelve months of trying, or six if you are over 35. Time matters and supplements are not a substitute for that appointment.

If you are pregnant, trying to conceive, or nursing, treat every entry here as ask-first rather than try-it.

See a doctor promptly for a breast lump, postmenopausal bleeding, bleeding between periods, sudden severe pelvic pain, visual changes with headache, or severe fatigue with dizziness, salt craving and skin darkening.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed, with some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions. It names the funding and the evidence tier, and lets you see both.

What outranks everything in this guide

For PCOS, lifestyle is first-line therapy, and the target is smaller than people think. A weight loss of 5 to 10% can significantly reduce reproductive, metabolic, and psychological symptoms and is recommended as first-line management, and the same source is candid that achieving it remains a real challenge for patients and clinicians (PMID 32814159).

Exercise has its own position statement, from Exercise and Sports Science Australia, which also asks clinicians to account for the significant psychological burden in PCOS, including weight stigma, and the high prevalence of comorbidities, when delivering exercise interventions (PMID 38960811).

I am quoting that second half deliberately. If you have been handed “just lose weight” as advice for a decade, the guideline literature is on your side about how inadequate that is as a plan.

For low testosterone in men under fifty, the cause is usually addressable. Prevalence estimates put biochemically low testosterone at 6 to 12% of men aged 40 to 49, commonly associated with obesity, metabolic syndrome, and sleep apnea (PMID 41127744). A scoping review of nonpharmacological interventions concludes that **proper physical exercise, sleep, and diet are indisputable approaches** because of their general health benefits, while nutraceuticals rest on weaker evidence and should be personalized by a physician or dietitian (PMID 35810031).

Sleep apnea in particular deserves a mention. It suppresses testosterone, it is common in exactly this group, and no supplement substitutes for treating it.

Key 3 in practice: the stress axis, and why it is not a supplement problem

Key 3 of the ebook covers the HPA axis and sex hormones under chronic stress. It is established physiology, and it is the key most often converted into a product it should not be.

What not to buy. Glandular and adaptogen protocols sold on the promise of fixing “adrenal fatigue.” That diagnosis is not recognized, and the lab guide covers why. Adrenal insufficiency, by contrast, is a genuine and serious diagnosis, and it is not found with a saliva kit; that one belongs with a physician.

On the testing itself. I do run four-point diurnal cortisol, salivary, in my practice. It is useful for the shape of the curve across a day, which is a different question from diagnosing a named adrenal condition, and it is worth separating those two things because the marketing in this space deliberately blurs them.

What is worth doing, and it is free. The exercise position statement in PCOS specifically asks clinicians to account for the significant psychological burden, including weight stigma, and the high prevalence of comorbidities when delivering interventions (PMID 38960811). That is guideline-level acknowledgement that the psychological load in this condition is part of the clinical picture rather than a side issue.

So treat load as a variable you manage rather than a panel you measure:

- **Five to ten minutes daily of slow breathing**, longer exhale than inhale. Consistency beats technique, and it costs nothing.
- **Protect sleep**, seven to nine hours on a fixed schedule. Short sleep worsens insulin resistance directly, which puts it upstream of Key 2.
- **Do not stack under-eating on top of hard training.** In this category that combination suppresses the reproductive axis, which is the opposite of the goal.
- **Name the stigma if it applies.** If you have spent years being handed weight advice instead of help, that is a load in itself, and it is worth saying out loud to a clinician who will listen.

Sourcing note: the breathing and sleep practices above are clinical reasoning rather than findings from a hormone-specific trial, and no citation is attached to them for that reason. They are included because they are free, carry no interaction risk, and act on a key the ebook identifies.

How to read this guide

Test first. The lab guide covers what. Fasting insulin is the one most often left off and it changes what you do here.

Give it three to six months. Cycle-related outcomes need cycles to measure, so a six-week trial answers nothing.

One change at a time.

Tier 1: inositol, for PCOS

This is the best-supported supplement in the whole kit, and it still needs its caveats stated.

What the research shows. An umbrella review synthesizing meta-analyses of randomized controlled trials assessed inositol in PCOS and found supporting evidence, while stating plainly that **outcomes based on low-quality evidence require cautious interpretation and should not solely guide clinical decisions**, and calling for larger, rigorous trials (PMID 41757236).

A systematic review and meta-analysis of combination therapy found that metformin plus inositol was associated with cycle regularization and reductions in hirsutism and the LH to FSH ratio compared with metformin alone, while noting that further studies are needed to clarify the true benefits (PMID 39331347).

A network meta-analysis comparing GLP-1 receptor agonists, metformin, and inositol on anthropometric and metabolic outcomes concluded that **no claims of overall superiority in PCOS management can be made** from that analysis, and called for long-term head-to-head trials across a broader range of outcomes (PMID 42490840). That paper carries a published correction, which is not a retraction.

And a useful nuance. A systematic review and meta-analysis found that **metabolic phenotype modifies the endocrine response** to inositol, meaning who you are metabolically may predict whether it helps your androgen picture. The authors describe this as a hypothesis-generating framework needing adequately powered, phenotype-stratified trials with clinically meaningful outcomes including standardized hirsutism and ovulatory measures (PMID 41947399).

How I read all that. Inositol is reasonable to try in PCOS, especially in the insulin-resistant phenotype, it is well tolerated, and the honest evidence grade is moderate at best with the umbrella review's own caution attached. It is not established as better than metformin, and the network meta-analysis says so.

Typical studied range. Trials most often use myo-inositol at around 4 g daily, frequently in a 40 to 1 ratio with D-chiro-inositol, taken in divided doses.

Cautions. Generally well tolerated, with mild gastrointestinal upset at higher doses. **If you take metformin, do not swap it for inositol on the strength of this guide.** The combination data above is about adding, not replacing, and that is a prescriber decision.

Tier 2: N-acetylcysteine, for PCOS

What the research shows. A systematic review and meta-analysis found NAC had a major effect on progesterone, endometrial thickness, and LH levels in women with PCOS, while finding **no significant differences in estradiol, SHBG, or FSH**, and concluding it may be a potential treatment option (PMID 39861414). A separate meta-analysis found NAC supplementation decreased total testosterone and increased FSH, suggesting it might improve reproductive system function in PCOS (PMID 36597797).

How to read it. Two meta-analyses, both positive on some hormonal measures and explicitly null on others. Note what the outcomes are: these are biochemical and ultrasound measures rather than “you will feel better,” and the two reviews do not agree on FSH direction, which is worth knowing.

Typical studied range. Trials commonly use 1.8 g daily in divided doses.

Cautions. Nausea is the common complaint. NAC can affect blood clotting at higher doses, which matters with anticoagulants and before surgery. Ask first in asthma.

Tier 3: correct what is measurably low, and ask first about the rest

Vitamin D, B12, ferritin, and magnesium. Test, correct a real shortfall, retest. In this population ferritin matters particularly for women with heavy cycles, where the threshold argument goes further than most reference ranges suggest.

Zinc and magnesium for testosterone appear in every men’s supplement aimed at this. The honest grade comes from the scoping review above: nutraceuticals for testosterone rest on weaker evidence than exercise, sleep, and diet, and should be personalized rather than taken speculatively (PMID 35810031). Correct a measured deficiency. Do not expect a supplement to raise normal testosterone.

Ask your prescriber first about these

- **Anything while on birth control, TRT, metformin, or thyroid medication**, which is most readers of this guide.
- **Anything at all if you are trying to conceive, pregnant, or nursing.**
- **Berberine**, which has real metabolic activity and real drug interactions, and which behaves more like a medication than a vitamin.
- **DHEA**, which is a hormone rather than a nutrient, is not benign, and should not be self-started.

What I deliberately left out, and why

- **DHEA and pregnenolone bought over the counter.** These are hormones. Self-prescribing them without testing and monitoring is not the same category as taking magnesium, and the androgenic side effects in women are real.
- **“Adrenal support” glandulars and cortisol protocols.** Adrenal fatigue is not a recognized diagnosis, and the panels sold to diagnose it are covered in the lab guide.
- **Estrogen “detox” protocols built on urine metabolite ratios.** The measurements themselves are analytically sound, and the lab guide covers that evidence fairly. What has not been shown is that steering supplements by those ratios changes outcomes, so I am not going to sell a protocol aimed at correcting a ratio.
- **Herbal blends marketed to “balance hormones.”** That phrase does not describe a mechanism, and the blends typically contain several actives at doses below anything studied.
- **Anything sold as a fertility treatment.** If conception is the goal, the appointment is the intervention.
- **Testosterone boosters generally.** The evidence grade is weak (PMID 35810031), and in men under fifty the cause is usually weight, sleep, or apnea, which a capsule does not touch.

Telling you what to skip is part of what you paid for.

Running a fair trial

Months 1 and 2, baseline. Track cycles, symptoms, and the two or three things that actually bother you most. Get the labs from the lab guide.

Months 3 to 8, one intervention. For PCOS with insulin resistance, inositol alongside the diet and exercise work is the reasonable first move.

Month 9, judge it. Compare cycles rather than days. Cycle regularity, hirsutism, and how you feel are the outcomes that matter, and a repeat panel is secondary.

Realistic expectations

What is realistic. More regular cycles over three to six months. Gradual improvement in metabolic markers. Slow improvement in hirsutism, which responds over many months because hair growth is slow.

What is not realistic. A supplement resolving PCOS, which is a long-term condition managed rather than cured. A capsule raising testosterone that is low because of untreated apnea. Any of this replacing a fertility evaluation.

Timeline. Three to six months minimum, judged in cycles.

When to seek professional care

Promptly. Breast lump, postmenopausal bleeding, bleeding between periods or after intercourse, sudden severe pelvic pain, visual changes with headache, or severe fatigue with dizziness, low blood pressure, salt craving and skin darkening.

Soon. No periods for three months outside pregnancy, rapidly worsening hirsutism or hair loss, milk discharge from the breast, or twelve months of trying to conceive, six if over 35.

Where to find quality versions of these

Nothing here names a brand, on purpose.

Look for third-party testing. NSF, USP Verified, or equivalent.

Match the form and ratio to the trials. Myo-inositol at around 4 g daily, often with D-chiro-inositol in a 40 to 1 ratio, is what the literature used.

Read the whole panel. “Hormone balance” blends frequently contain DHEA or botanicals with hormonal activity that are not on the front of the label.

If you would rather not evaluate all of that yourself, I keep a curated dispensary of professional-grade products at Root Health, screened for the standards above.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the first-line interventions in this category, movement, sleep, and food, earn us nothing.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your cycle history, your medication list, and whether conception is a goal, because that last one changes the whole approach.

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If you would rather start on your own, track first:

Free symptom tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your Hormone Labs** covers what serum, saliva, and urine testing can and cannot tell you, and **The Hormone Diet Guide** covers the insulin work that sits underneath most of this.

References

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